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Practical Aspects of Implementing an Integrated System for the Sale, Installation and Monitoring of Photovoltaic Systems

Małgorzata Oleś-Filiks^{a,*}, Robert Waszkowski^b

^aUniversity of Warsaw, Faculty of Management, Szurmowa 3, 02-678 Warsaw, Poland

^bMilitary University of Technology, Faculty of Cybernetics, Kaliskiego 2, 00-908 Warsaw, Poland

Abstract

The purpose of this study is to present a designed and implemented an integrated system for the sale, installation and monitoring of photovoltaic systems in a company that is one of the leaders on the Polish renewable energy market. The study also conducted a literature review on photovoltaic systems and green jobs. The basic architecture of the system envisages the use of a number of components responsible for: data preparation in the sales module; implementation in the installation module; supervision and monitoring of actions by management in the monitoring module. The integrated approach presented: increases efficiency through streamlined workflows; improves transparency through a central source of information; enables efficient management of resources; while monitoring the progress of management activities. In addition, the implementation of the integrated system has enabled real-time synchronisation of work between the sales department and the investment implementation and maintenance department.

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1. Introduction

Recently, there has been a remarkable increase in the global demand for renewable energy sources, resulting in a significant rise in the use of solar energy solutions [1]. In a world where energy sources are rapidly becoming scarce [2], and where energy consumption is increasing over time, causing increasing environmental pollution [3, 4], solar

* Corresponding author. Tel.: +48 22 55 34 129, E-mail address: moles@wz.uw.edu.pl

energy is presented as an inexhaustible solution to meet our energy needs [5]. Various studies show that solar energy is one of the most widely used renewable energy sources today [6]. It should be noted that this energy source has a minimal impact on the environment, which is highly relevant for the care of the planet [7], [8].

Solar panels play a critical role in solar energy generation, capturing sunlight and converting it into environmentally friendly electricity [9]. The solar panel is the most appropriate device to take advantage of this energy source [10], since it uses photovoltaic solar technology, converting sunlight into direct current through solar cells [11], which feeds electrical energy to various devices [12].

Photovoltaic (PV) systems have gained prominence as a sustainable and environmentally friendly source of electrical energy. Their use is key to reducing our dependence on fossil fuels and mitigating the adverse effects of climate change. Fossil fuels are the main cause of global warming, but using renewable energy resources is able to minimize greenhouse gas emissions, which are the main sources of global warming [13-14]. However, maintaining optimal solar panel performance and maximising energy output requires continuous monitoring and data analysis [15].

The assessment of PV hosting capacity in distribution networks is currently an emerging technological hotspot, widely used in the design of distribution networks with a high proportion of photovoltaic access. Rooftop PV is generally installed on the roofs of buildings. Since the area of buildings varies in different electricity supply areas, considering the potential estimation of rooftop PV can provide more accurate evaluation results of PV hosting capacity, which helps to analyse the potential operational risks of the distribution system after large-scale rooftop PV grid connection [16].

Most of the available studies analyse technological scenarios that combine other renewable energies (e.g. wind power) [17] or examine future targets and guidelines [18]. Some studies examine the issue from the perspective of the future sustainability of energy diversification, in particular with regard to its economic and environmental aspects [19] in terms of photovoltaic energy production in two European cities, Berlin and Rome [20]. Also relevant is the study that uses a spatial auto-regressive model to assess the geographical distribution of photovoltaic production capacity [21]. There is also work that is noteworthy for its consideration of future changes in energy production, taking into account improvements in the efficiency of photovoltaic cells due to technological progress [22]. There are also studies that model changes in the energy production of photovoltaic systems due to climate change, based not on climate scenarios but on mathematical probability forecasts [23] or machine learning [24-26], and others that examine photovoltaic energy production in conjunction with other renewable energies in shaping future energy security [27].

Various studies have shown that households account for approximately 40% of global energy consumption and play a crucial role in the energy market. In addition, residential energy demand is expected to continue to grow globally over the coming decades [28]. In different regions of the world, the percentage of total consumption accounted for by residential buildings varies, ranging from an average of 20% in developed countries to more than 35% in developing countries. Therefore, residential energy demand is significant both in terms of its current size and its potential for growth [29]. Considering that the demand for electrical energy consumption in the residential and industrial sectors has increased over the years [30].

As industrialisation has progressed, so has the demand for energy. Traditional fossil fuels are becoming depleted and their use is having a negative impact on the global climate [31-33]. Research suggests that by 2100 the average global temperature will have risen by more than 5°C compared to pre-industrial levels [34]. With the ongoing development and progress of industrialisation, the demand for energy continues to grow, leading to the exploration and use of renewable green energy sources. Solar energy, which is derived from the sun's radiation [35], can be converted into electrical or thermal energy using solar panels. It has applications in electricity generation, heating and various other energy needs. Solar energy is a clean and environmentally friendly source with huge potential. It can reduce dependence on fossil fuels, reduce greenhouse gas emissions and contribute to achieving sustainable energy goals. Solar energy is converted into electricity by photovoltaic (PV) cells, the output of which depends on the amount of solar energy collected [36]. The principle behind the process is that when sunlight strikes the solar panel, photons excite electrons in the semiconductor, causing them to move and generate an electric current.

The performance of a PV, PV/T and PV/wind photovoltaic system depends on the climatic conditions, the equipment used and the configuration of the system. This performance can be measured as the ratio between the actual output of the photovoltaic solar system and the expected values. Measurement by means of sensors is essential

for the proper operation and maintenance of the photovoltaic system. Performance values are determined using PV monitoring systems consisting of a data recorder and, in most cases, a meteorological instrument [37].

With the help of small-scale PV systems, the dependence on the national grid is reduced and the monthly electricity bill is lowered. Sunlight is not a reliable energy resource as there are some following challenges associated with it. In recent years, with the implementation of nationwide photovoltaic policies and price subsidies for photovoltaic products, rooftop photovoltaic has attracted more and more attention. Large-scale grid-connected rooftop PV has changed the characteristics of traditional distribution networks, and the power flow of the distribution network has shifted from "one-way" to "two-way". High penetration of PV will have adverse effects on the safe and stable operation of existing distribution systems, such as voltage exceeding limits, increased risk of equipment thermal stability and relay protection failure [13-14].

The term "green jobs" has become more widely used in the literature in recent years, but the approach to areas offering such opportunities and specific skills varies between published papers or from one country to another. Meanwhile, the lack of a widely accepted definition raises several issues, including the precise highlighting of areas and the exact number of such jobs. Valero et al. have shown that a top-down approach that uses a narrow definition of green jobs, considering only those industries or activities that are directly relevant to decarbonisation, leads to an estimate of green jobs below 5% of employment in the US or European economies. The same study emphasises that a "bottom-up" definition, which includes jobs directly or indirectly related to decarbonisation, can yield much higher percentages [38].

Another approach is to classify green jobs as either outcomes or processes. The first perspective is that these jobs produce green goods or services, such as green buildings, clean transport, solar water heaters, or photovoltaic systems. The second is that green jobs contribute to greener processes, for example by reducing water consumption, controlling air pollution or improving recycling services [39]. In line with this approach, the centralised system for managing the sale, installation and monitoring of photovoltaic systems fits perfectly with the theme of "green jobs".

Access to energy resources and the quality and reliability of the energy system are important not only for the ability of individual economic sectors to develop, but also for the quality of life of individuals or households [40].

There is a right relationship between the labour market and the environment. A green labour market is a way of doing business that aims to reduce the consumption of natural resources, produce less waste, reduce greenhouse gas emissions and, above all, eliminate social inequalities. The increasing number of green jobs and the growing popularity of green products and services are signs of the growing influence of the green economy on economic practice. [41].

Taking into account these references and conclusions, an integrated system for the sale, installation and monitoring of photovoltaic systems was developed and implemented - a system dedicated to the photovoltaic industry, as described in this article.

When implementing projects and investments (for both private and business customers) in the form of resource-intensive photovoltaic parks, the company needs a reliable IT solution that can support more than 5,000 users in the processes of sales, design, installation and monitoring of the installed systems.

The basic architecture of the centralised system for managing the sale, installation and monitoring of photovoltaic systems involves the use of a number of components responsible for: data preparation in the sales module; implementation in the installation module; supervision and monitoring of the process by management in the monitoring module. In addition, the system allows the configuration of specialised dedicated modules: quotation calculator by the customer; installation matching with drone footage; electronic signatures with integrated CRM; electronic payments; mechanism for adding services and additional costs to calculations; adding interfaces to external CRM, ERP systems and databases. The solution designed and implemented was based on dedicated modules and the Aurea BPM platform. The Aurea BPM platform enables the dynamic design of the company's business processes.

The actions performed by individual user groups within modules and processes have a direct impact on the related components and processes. Mutual communication within the tasks performed allows: synchronisation of work between the sales department and the investment implementation and maintenance department; continuous reporting on sales progress; acceleration of the flow of information and improvement of the quality of customer service; monitoring of completed installations in the maintenance phase. Thanks to the use of the latest technology, users who use the system on multiple devices (information kiosks, tablets, smartphones and computers) have access

to up-to-date data. The system for managing the sale, installation and monitoring of photovoltaic systems records every action and decision taken by the parties involved in a process history with an exact time stamp and a record of who carried out the action.

No studies were found in the literature describing the practical aspects of implementing an integrated system for the sale, installation and monitoring of photovoltaic systems, indicating that there is a research gap in this area.

The result of the implementation was presented in the form of business process diagrams in BPMN notation. The system was implemented in a company that is one of the leaders in the Polish renewable energy market. The system provided a single place for information and knowledge about sales to customers and accelerated the flow of information, which allowed improving the quality of customer service. For managers, the system provides excellent access to management data and the ability to set key performance indicators.

2. Assumptions for an integrated system for the sale, installation and monitoring of photovoltaic systems

Prior to the start of the work, the key assumptions for an integrated system to manage the sale, installation and monitoring of PV systems were identified and described. The conceptual work was carried out following in-depth interviews with managers and owners of a company that is a leader in the photovoltaic market in Poland. During the conceptual work, the following system assumptions were identified:

- optimise the use of automation for the sale, quotation and monitoring of photovoltaic systems,
- increase sales efficiency by reducing the time needed to prepare quotations,
- support process management to increase sales and maintenance efficiency,
- centralisation of activities,
- continuous reporting of sales activities,
- cooperation of the modules with the storage system,
- calendar for synchronising appointments with the customer,
- integration with internal financial, workflow, sales and warehouse systems,
- monitoring of completed installations during the maintenance phase,
- management of a repository for contracts and documents,
- electronic signatures (also on tablets),
- integration with electronic transfers.

The planning and preparation process for the implementation of the system included activities such as:

- planning and preparing the terms of reference,
- working with the project approach,
- carrying out a comprehensive business analysis with a view to building a new system,
- the need to formulate functional assumptions and project objectives,
- integration with existing infrastructure,
- preparing staff and process owners for the changes resulting from the implementation of the new system,
- identifying priorities such as progress monitoring, automation and building a reporting system,
- conduct end-user meetings and training.

3. System architecture and components

The basic architecture of the centralised system for the sale, installation and monitoring of photovoltaic systems consists of the following modules, see Fig. 1.:

- sales module,
- installation module,
- monitoring module.

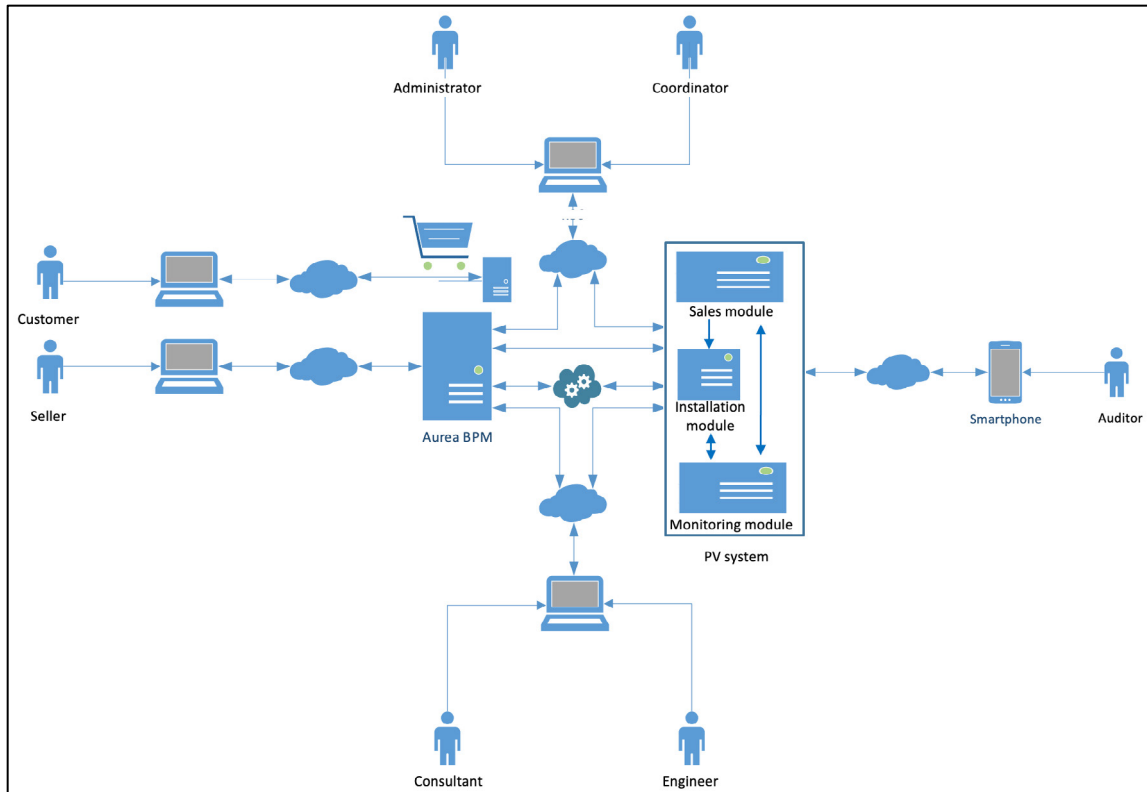


Fig. 1. The core architecture of integrated system for sales, installation and monitoring of photovoltaic systems.

3.1. Sales module

The sales module is designed to manage between a few dozen and a few hundred of the operator's salespeople, creating offers for external customers in the B2B and B2C model. The assumption was to work post CRM, on a set of customers interested in obtaining offers for the sale, installation, maintenance of photovoltaic sets. The implemented system guides the user through the entire advanced process of submitting a commercial offer. A set of customers is created from email enquiries, website forms, information kiosks or data entered manually by sales staff. An initial investment calculation is made based on basic data: address, customer type, power requirement, panel type. Once the location has been entered, the implemented system can refer to an automatically loaded solar map, taking into account the roof pitch, orientation to the south, number of modules and expected nominal power, see Fig. 2. and Fig. 3. There is an option to create quotes with the possibility of inserting additional customer conditions. The customer receives the quote in PDF format, which is sent for approval and signed by the quote provider. The next step is to send the quotation by email or on paper. This is followed by a site visit - the consultant writes down the details; selection of the energy operator; writing down the meters; description of the conditions for installing the connection; verification of who the contract is signed for; there is an option to add photos of the roof with the possibility of using a drone, see Fig. 4. This is followed by the sending of post-audit data for multi-level technical verification, the sending of information to the dealer, and the generation of the installation wiring diagram. The next step is to approve the final offer and send the material to the customer, consisting of the audit report, the contract and the offer. The whole thing can be signed by the customer - manually or digitally. Upon receipt, the consultant accepts the order in the system on behalf of the client as ready for execution. Waiting for advance payment - can be made via the electronic payment module or by bank transfer to an individual account number. An invoice is automatically generated for the client. The project is then handed over to the installation team for implementation. The contract configurator assists the customer in the purchase

process and reduces the possibility of errors. This can be important when extending the offer to other areas such as recuperation, air conditioning or heat pumps.

Roof information

ADD A ROOF +

Number of panels on the roof 1	The maximum possible number of panels on the roof 0	inclination angle [°] 6	Roof orientation [°] 12
Module mounting location Sloping roof	Type of roofing Trapezoidal metal sheet	Type of mounting structure Trapezoidal bridge - black (33cm)	A different type of coverage
Tile width/distance between bridges/seams [cm]		Shading or obstructions NO	
Height of the roof tile/trapezoidal bridge/seam [cm]		Type of roof structure	
Does the customer have spare tiles? ☒ Yes ☐ NO ☐ Not applicable		The distance between the rafters The installation distance from the edge The height of the building from the ground to the eaves	
Roof dimensions [m]:	Width 12	Length 20	2 2 3
Comments brak			

SHOW PRODUCTIVITY FOR THE ROOF

GO TO THE VISUALIZATION

ADD/EDIT ROOF PHOTO

REMOVE THE ROOF

Fig. 2. Roof information.

Calculation of productivity for the roof

Annual [kWh]
353.22

January [kWh] 5.79	February [kWh] 10.74	March [kWh] 26.7	April [kWh] 43.24	May [kWh] 53.36	June [kWh] 55.58
July [kWh] 51.31	August [kWh] 45.21	September [kWh] 30.8	October [kWh] 17.54	November [kWh] 7.47	December [kWh] 5.5

COME BACK

Fig. 3. Calculation of productivity for the roof.



Fig. 4. The visualization of the roof.

3.2 Installation module

The installation module responsible for automatically calculating the need for the elements required for installation according to the rules defined in the system. This module is the main element of the system, which enables the creation, management and tracking of initiatives registered in the system by users. The algorithms implemented in this module automatically send orders to the warehouse system based on the contract data. This is followed by the planning of the installation date with the support of graphic tools - Gantt charts. Thanks to the possibility of defining approval paths, the activities created can be checked and approved by the required persons and then sent for execution. Thanks to special forms, users can describe and categorise the activities entered into the system, which greatly facilitates their further evaluation and verification. The next step is to set an acceptance date and send the information to the customer, and acceptance of the installation can take place directly in the system. The final step is commissioning and linking to the plant performance monitoring system.

3.3. Monitoring module

The monitoring module is integrated with the modules described above and is based on the data collected in them. In this module it is possible to: efficiently manage activities; monitor the progress of work carried out within them; supervise and monitor the progress of activities from the management side; efficiently manage the sales team; design and implement dynamic reports; view management information at the level of what is currently happening in sales. The monitoring module provides comprehensive monitoring and control of installed equipment. Automatic reporting of system failures allows the operator to react quickly. This module also allows comparative analysis with other installations in the area in terms of efficiency and correct operation. This module is responsible for managing the budget and all the work associated with the activities by creating a schedule of activities, as well as controlling expenses by attaching specific expenses to previously created activities and schedule activities.

In addition, the system allows the configuration of specialised dedicated modules:

- a quotation calculator by the client,
- adaptation of the system to drone images,
- electronic signatures with integrated CRM,
- electronic payments,
- mechanism for adding services and additional costs to the calculations (e.g. sale of heat pumps, air conditioning, energy storage),
- add interfaces to external CRM, ERP systems, databases,
- mechanism for adding services and additional costs to the calculations - e.g. adding costs for excavation, use of lifting equipment, additional sale of heat pumps, air conditioning, energy storage,
- option to sign on a tablet.

4. Challenges and conclusions from the implementation

The design work identified the following challenges to implementing a system to support the sale, installation and monitoring of PV systems:

- the design of specific forms for PV system site audits,
- the need to combine traditional statutory activities with new ways of working,
- modelling PV system equipment selection calculators,
- running multiple businesses in parallel with a complex organisational structure,
- simultaneous operation of several thousand sales advisors,
- limited financial resources; parallel project work and ongoing tasks,
- modelling a multi-level organisational structure taking into account hierarchies and regionalisation of users,
- implementing electronic workflows for processes that were previously performed in an analogue manner,
- providing advanced tools for creating templates for your own offers and contracts.

The implementation of an integrated system for the sale, installation and monitoring of photovoltaic systems - a system dedicated to the photovoltaic industry - has had a significant impact on the performance of current tasks:

- reducing waste, social responsibility through continuous learning and adopting environmental objectives and strategies,
- implementation contributes to sustainable development, leading to a green economy and green jobs,
- increased sales efficiency through a significant reduction in the time needed to prepare quotations, giving sales staff more time to win customers,
- maintaining quality standards for quotations,
- easier communication with the customer, eliminating missing documents and internal and external correspondence,
- easily adaptable work platform for new employees,
- synchronisation of work between sales and project management and maintenance departments,
- a dedicated quotation tool where, instead of a generic visualisation, you get a comparison of the installation with drone footage,
- collaboration of the configurator with the warehouse system,
- optimisation of stock rotation,
- calendar for synchronisation with customer, contract and installation dates,
- monitoring of completed installations during the maintenance phase,
- ERP integration for contract invoicing,
- contract and document repository,
- electronic signatures (also on tablets),
- integration of electronic transfers,
- effective reporting to support management activities,
- elimination of internal supplier competition through clear customer identification.

The implementation of an integrated system for managing the sale, installation and monitoring of photovoltaic systems has achieved the following objectives:

- development of staff responsible for environmental activities,
- synchronisation of the sales process with the warehouses, together with the functionality of blockers that show the actual status of the goods and block the possibility of selling goods that are not physically present,
- maintain a single platform of information on current issues available to authorised participants in the process,
- allows comprehensive budget management, process control, integrated reporting system to be handled in one place,
- reduction of routine and unnecessary formalities in favour of automation,
- possibility of handling processes electronically (accessibility, remote working, registers, reports),
- easier management and implementation of projects,
- increase the efficiency of work organization,
- optimisation of budget utilisation through process management,
- easier monitoring of work progress,
- faster and easier reporting of ongoing activities,
- integration with an external system, including a smooth flow of information and data between different areas,
- automatically maintained records of documents,
- ongoing reporting on sales progress.

5. Conclusions

The achievement of the implementation goal was made possible by the active involvement of users in the planning phase, which allowed the specific characteristics of the industry and the needs of photovoltaic companies to be taken into account. Furthermore, the ability to modify and adapt the platform to individual needs is essential for photovoltaic implementations. At this point, it is also worth mentioning that the planning of the implementation objectives; the definition of the functional scope is fundamental and is a step that cannot be omitted due to the flexibility of the implementation and the adaptability of the system, which is essential in a dynamically changing environment.

The following advantages of the implemented system of sales, installation and monitoring of photovoltaic systems were highlighted: a single location for information and knowledge on sales to the customer; elimination of lost documents; acceleration of the flow of information and increase in the quality of customer service; updating of information on offer prices with a distributed sales team; A unified and flexible reporting system in areas defined by the operator; an effective tool for modelling and documenting the company's processes; easy training and onboarding of new employees; excellent access to management data and the ability to set KPIs; automation of document handling, data validation and reporting shortens decision-making time and improves communication between process participants; quick access to data and manipulation according to defined rights. A key achievement of the implementation was also the creation of a reporting system for internal use and group work. Aurea BPM, as a low-code platform, allowed extensive involvement of process owners and users in the phase of agreeing the shape of the target application.

The system of sales, installation and monitoring of photovoltaic systems differs from systems of this class available on the market due to: organisation of the commission system for the operator's sales channels; a single pricing policy - the customer reports to different consultants of the same operator; intelligent cost simulation before installation, detailed designs based on photos and measurements taken by the drone, consideration of all factors of current energy consumption when calculating return on investment.

It is worth noting that the implementation of the system has made it possible to manage the financial department more effectively, including monitoring the implementation of approved projects, which has made it possible to estimate and simulate future activities. The implementation of the system has improved communication with suppliers and subcontractors in the preparation of contracts and documentation, as well as in the execution and settlement of the tasks assigned to them. Considering the importance of remote working for the organisation in the face of today's epidemic threats, the implemented system for managing the sale, installation and monitoring of photovoltaic systems allowed access to data from any location through secure and state-of-the-art web interfaces.

No studies were found in the literature on the practical aspects of implementing an integrated system for the sale, installation and monitoring of photovoltaic systems.

The completed study had its limitations - both the design and implementation work was carried out on the Polish market. Closely related to these limitations is the need for further research that could be extended to European markets.

Undoubtedly, the research carried out has an impact on the scientific aspect in the area described above - to date, no system for the sale, installation and monitoring of photovoltaic systems - a system dedicated to the photovoltaic industry - has been designed and implemented successfully and on such a large scale on the Polish market. The description of the implementation can undoubtedly serve as a model and core for implementations in this field.

References

- [1] Nogueira, Duarte and Machado (2022) "Review of the recent advances in solar assisted heat pumps integrated with renewable energy sources", *Brazilian Congress of Thermal Sciences and Engineering*.
- [2] Mattick, Williams and Allenby (2010) "Historical trends in global energy consumption", *IEEE Technology and Society Magazine*, vol. 29, no. 3: 22–30.
- [3] Ananthu, Kashappa and Venkateshkumar (2022) "Artificial intelligent controller-based energy management system for grid integration of PV and energy storage devices", *Indonesian Journal of Electrical Engineering and Computer Science (IJECS)*, vol. 26, no. 2: 617–628.
- [4] Saleem, Iqbal, Mehmood, Panjwani, Mangi, and Larik (2019) "The effect of environmental changes on the efficiency of the PV system", *Indonesian Journal of Electrical Engineering and Computer Science (IJECS)*, vol. 18, no. 1: 558–564.
- [5] Maka, Alabid (2022) "Solar energy technology and its roles in sustainable development", *Clean Energy*, vol. 6, no. 3: 476–483.
- [6] Susrutha, Giribabu and Singh (2015) "Recent advances in flexible perovskite solar cells", *Chemical Communications*, vol. 51, no. 79: 14696–14707.

- [7] Stoppato (2008) “Life cycle assessment of photovoltaic electricity generation”, *Energy*, vol. 33, no. 2: 224–232.
- [8] Hernandez et al. (2014) “Environmental impacts of utility-scale solar energy”, *Renewable and Sustainable Energy Reviews*, vol. 29: 766–779.
- [9] Shaikh (2017) “A review paper on electricity generation from solar energy”, *International Journal for Research in Applied Science and Engineering Technology*, vol. V, no. IX.
- [10] Smestad et al. (2008) “Reporting solar cell efficiencies in solar energy materials and solar cells”, *Solar Energy Materials and Solar Cells*, vol. 92, no. 4: 371–373.
- [11] Al-Ezzi, Ansari (2022) “Photovoltaic solar cells: a review”, *Applied System Innovation*, vol. 5, no. 4: 67.
- [12] Aneke, Wang (2016) “Energy storage technologies and real life applications – a state of the art review”, *Applied Energy*, vol.179: 350–377.
- [13] Andresen, Buticchi and Liserre (2016) “Thermal stress analysis and MPPT optimization of photovoltaic systems”, *IEEE Trans. Ind. Electron*, 63: 4889–4898.
- [14] Zhan, Wang, Wang, Yang, Zhang, Wu and Chen (2016) “Relay protection coordination integrated optimal placement and sizing of distributed generation sources in distribution networks”, *IEEE Trans. Smart Grid*, 7: 55–65.
- [15] Olorunfemi, Ogbolumani, and Nwulu (2022) “Solar panels dirt monitoring and cleaning for performance improvement: a systematic review on smart systems”, *Sustainability (Switzerland)*, vol. 14, no. 17.
- [16] Zhong, Zhang, Chen, Zhang, Zhou, Zhu, Wang, Lü and Yan (2021) “A city-scale estimation of rooftop solar photovoltaic potential based on deep learning”, *Appl. Energy*, 298, 117132.
- [17] Campos, Csontos and Munkácsy (2023) “Electricity scenarios for Hungary: Possible role of wind and solar resources in the energy transition”, *Energy*, 278.
- [18] Atsu, Seres and Farkas (2021) “The state of solar PV and performance analysis of different PV technologies grid-connected installations in Hungary”, *Renewable and Sustainable Energy Reviews*, 141.
- [19] Madaleno Ahmed Doğan Javeed and Vasa (2023) “The aptness of import-led growth hypothesis for sustainable development in South Asia: Do energy utilization and natural resources matter?”, *Resources Policy*.
- [20] Baglivo, Congedo and Mazzeo (2023) “Scenarios for urban resilience—perspective on climate change resilience at the end of the 21st century of a photovoltaic-powered mixed-use energy community in two European capitals”, *Adapting the Built Environment for Climate Change: Design Principles for Climate Emergencies*, Woodhead Publishing, 37–52.
- [21] Copiello, Grillenzoni (2017) “Solar Photovoltaic Energy and Its Spatial Dependence”, *Energy Procedia*, 141: 86–90.
- [22] Oka, Mizutani and Ashina (2020) “Climate change impacts on potential solar energy production: A study case in Fukushima, Japan”, *Renewable Energy*, 153: 249–260.
- [23] Abdulai, Gyamfi, Diawuo and Acheampong (2023) “Data analytics for prediction of solar PV power generation and system performance: A real case of Bui Solar Generating Station, Ghana”, *Scientific African*.
- [24] Ghimire, Deo, Casillas-Pérez and Salcedo-Sanz (2022) “Boosting solar radiation predictions with global climate models, observational predictors and hybrid deep-machine learning algorithms”, *Applied Energy*.
- [25] Nwokolo, Obiwulu and Ogbulezie (2023) “Machine learning and analytical model hybridization to assess the impact of climate change on solar PV energy production”, *Physics and Chemistry of the Earth*.
- [26] Chukwujindu, Nwokolo, Ogbulezie and Ummunnakwe Obiwulu (2022) “Impacts of climate change and meteo-solar parameters on photosynthetically active radiation prediction using hybrid machine learning with Physics-based models”, *Advances in Space Research*, 70(11): 3614–3637.
- [27] Svazas, Navickas, Paskevicius, Bilan and Vasa (2023) “Renewable Energy versus Energy Security: The Impact of In-Novation on the Economy”, *Rynek Energii*, 164(1): 60–71.
- [28] Nejat, Jomehzadeh, Taheri, Gohari and Mueh (2015) “A global review of energy consumption, CO2 emissions and policy in the residential sector (with an overview of the top ten CO2 emitting countries)”, *Renewable and Sustainable Energy Reviews*, vol. 43: 843–862.
- [29] Yau, Hasbi (2013) “A review of climate change impacts on commercial buildings and their technical services in the tropics”, *Renewable and Sustainable Energy Reviews*, vol. 18: 430–441.
- [30] Kelly (2012) “Sustainability at home: policy measures for energy-efficient appliances”, *Renewable and Sustainable Energy Reviews*, vol. 16, no. 9: 6851–6860.
- [31] Zhang, Zeng, Li and Yang (2020) “Impact of PDO and AMO on interdecadal variability in extreme high temperatures in North China over the most recent 40-year period”, *Clim. Dyn.* 54, 3003–3020.
- [32] King, Karoly and Henley (2017) “Australian climate extremes at 1.5 °C and 2 °C of global warming”, *Nat. Clim Chang*, 7: 412–416.
- [33] Jacobs, Buizer and Moser (2016) “The third US national climate assessment: Innovations in science and engagement”, *Clim. Chang*, 135: 1–7.
- [34] Zhang, Zeng, Yang and Jiang (2021) “Future Changes in Extreme High Temperature over China at 1.5 °C–5 °C Global Warming Based on CMIP6 Simulations”, *Adv. Atmos. Sci.* 38:253–267.
- [35] Racharla, Rajan (2017) “Solar tracking system—A review”, *Int. J. Sustain. Eng.*, 10: 72–81.
- [36] Yao, Hu, Gao, Yang and Du (2014) “Multipurpose dual-axis solar tracker with two tracking strategies”, *Renew. Energy*, 72: 88–98.
- [37] Allafi, Iqbal (2018) “Low-cost SCADA system using arduino and reliance SCADA for a stand-alone photovoltaic system”, *Journal of Solar Energy*, art. no. 3140309.
- [38] Sulich, Rutkowska, Popławski (2020) “Green Jobs, Definitional Issues, and the Employment of Young People”, *An Analysis of Three European Union Countries*, *J. Environ. Manag.*, 110314: 262.
- [39] Ree (2019) “Promoting Green Jobs: Decent Work in the Transition to Low-Carbon”, *Green Economies. Int. Dev. Policy*, 11: 248–271.
- [40] Kozar, Sulich (2023) “Green Jobs in the Energy Sector”, *Energies* 2023, 16, 3171.
- [41] Sulich, Rutkowska (2021) „Zielony rynek pracy a zielona rewolucja”, Preprint.